

Second Harmonic Generation in Lithium Niobate on Insulator

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This paper reviews recent advancements in second harmonic generation (SHG) within nanostructures of lithium niobate on insulator (LNOI). SHG devices are classified into in-plane and out-of-plane configurations, highlighting their suitability for different pumping conditions. The in-plane devices, with their small mode volume and ultra-narrow linewidth, demonstrate high-efficiency SHG under continuous wave pumping, making them ideal for miniaturized on-chip integrated devices. In contrast, the vertical cavities are best suited for broadband pulsed lasers, facilitating the creation of compact ultra-thin devices. This review summarizes the recent strides made in enhancing SHG efficiency for different designs in LNOI platform and underscores the potential of LN micro- and nanostructures to fulfill the requirements of modern ultracompact nonlinear devices.

The nonlinear wave equation governing the electric fields of the fundamental frequency (FF) E_1 and the second harmonic (SH) E_2 is:^[16]

$$\nabla^2 E_j + \frac{n_j^2 \omega_j^2}{c^2} E_j = -\frac{\omega_j^2}{\epsilon_0 c^2} P_j^{(2)} \quad (1)$$

where ϵ_0 , c , and n are the permittivity, the light speed in vacuum, and the refractive index, respectively. The indices $j = 1$ and 2 correspond to the FF and SH components. The expressions for the second-order nonlinear polarization amplitudes are given by

$$\begin{cases} P_1^{(2)} = 2\epsilon_0 \chi^{(2)} : E_2 E_1^* \\ P_2^{(2)} = \epsilon_0 \chi^{(2)} : E_1 E_1 \end{cases} \quad (2)$$

1. Introduction

Second harmonic generation (SHG) is a fundamental nonlinear $\chi^{(2)}$ process that is critical for a variety of optical and photonic applications. SHG enables the conversion of light from one wavelength to exactly half of that wavelength.^[1,2] This effect is particularly useful for extending the spectral coverage of laser sources beyond the limitations of standard laser gain media. SHG is also promising for various optical signal processing tasks, including frequency mixing,^[3] signal amplification,^[4] and modulation.^[5–7] By leveraging SHG, complex functions can be performed, such as optical switching,^[8,9] routing,^[10] and waveform generation,^[11–14] which are essential for advanced optical communication systems.^[15]

where $P_2^{(2)}$ governs the SHG, and $\chi^{(2)}$ is the second-order nonlinear susceptibility of material. Meanwhile, $P_1^{(2)}$ describes the back-conversion process from SH to FF. Currently, there are different methods to characterize SHG efficiency.^[16,17] The most fundamental metric is the absolute SHG efficiency (η), is typically expressed as the ratio of the SH average power P_2 to the pump average power P_1 : $\eta = P_2/P_1$. However, since this efficiency depends on the input power P_1 , a normalized efficiency η_{norm} is commonly used to eliminate the influence of input power: $\eta_{\text{norm}} = P_2/P_1^2$. For pulsed light excitation, the peak intensity is commonly used to account for variations laser parameters such as fluence, pulse duration, repetition rate, and focused spot area.^[18] In this case, the intensity-normalized efficiency is defined as $\eta_{\text{norm},I} = \eta/I_{\text{pk}}$, where I_{pk} is the pump peak intensity.^[19] In addition, since SHG power accumulates along the propagation distance, a length-normalized efficiency $\eta_{\text{norm},L}$ has been introduced to remove dependence on the sample length. It is defined as: $\eta_{\text{norm},L} = P_2/(P_1^2 L^2)$. Efficient SHG traditionally requires strict phase-matching conditions to ensure constructive interference of the generated nonlinear optical signals and continuous power transfer from the pump to the signal.^[16,20,21] Phase matching is typically achieved through careful control of the refractive indices of interacting optical waves, via tuning angular orientation of the crystal or varying its temperature.^[22] Domain engineering also provides an effective method for achieving quasi-phase matching (QPM) by introducing periodic variations in the nonlinear coefficient of the materials (e.g., periodically poled ferroelectric crystal).^[23–27] To date, high-power, phase-matched, and long-interaction-length processes have been

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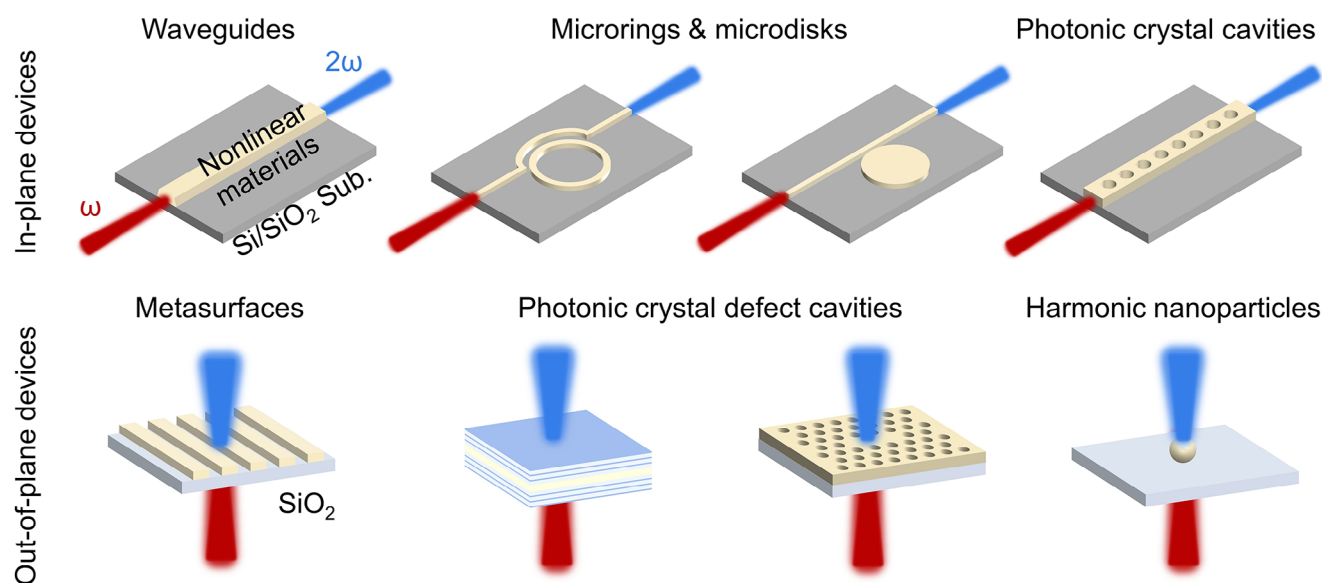


Figure 1. Two main classes of nonlinear LNOI devices. One is based on a in-plane design, such as waveguides, microrings, microdisks, and photonic crystal cavities. The other is out-of-plane vertical cavities, including metasurfaces, photonic crystal defect cavities, and harmonic nanoparticles.

achieved in bulk materials, leading to efficient second-order nonlinear optical processes.^[28,29]

Among the various materials under explored, lithium niobate (LiNbO_3 , LN) is widely regarded as an ideal candidate for such applications. The LN is a multifunctional crystal with many outstanding optical properties.^[30] It is transparent across a broad spectral range from ultraviolet to mid-infrared, and possesses a large $\chi^{(2)}$ nonlinear coefficient. Therefore, LN can operate SHG in a wide spectral range.^[31,32] To meet the future requirements of device miniaturization, the focus of optoelectronic research has shifted from bulk discrete optics to thin films based integrated photonics. This transition calls for thin film materials that exhibit key properties such as low light propagation loss, and efficient optical nonlinearities. The recent lithium niobate-on-insulator (LNOI) has emerged as a promising platform to address these challenges. The large refractive index contrast between the LN film and the transparent substrate (such as silica) leads to a well optical mode confinement and high optical intensity within the submicron-thin LN layer, which makes LNOI very attractive and competitive for nonlinear integrated photonics applications.

In this paper, we provide an extensive review of the latest advancements in LNOI-based SHG micro- and nanostructures. These devices can be primarily categorized into two types based on their configurations: in-plane and out-of-plane designs, as illustrated in **Figure 1**. In-plane devices operate within the same plane as the substrate or chip surface. Typical examples include waveguides, microrings, microdisks, and photonic crystal cavities.^[33] These structures play a crucial role in integrated photonics, not only for their scalability and compatibility with existing semiconductor fabrication technologies, but also for their ability to confine light within the LN core layer over a considerable propagation distance without diffraction. The planar nature of these devices facilitates efficient SHG through tight mode confinement and effective PM within the planar structure. Conversely, out-of-plane devices emit and manipulate SHG waves

along the out-of-plane direction. This category includes metasurfaces, vertical photonic crystal defect cavities, and harmonic nanoparticles. Out-of-plane configurations are advantageous for applications requiring light manipulation outside the plane of the substrate, such as in displays, optical data storage, and free-space optical communications. Through this review, we aim to provide a comprehensive introduction to the design principles, fabrication techniques, and performance metrics of both in-plane and out-of-plane SHG devices. By exploring these advancements, we hope to inspire further research and innovation in the field of nonlinear photonics, leveraging the unique properties of LNOI for enhanced optical functionalities.

2. Results

2.1. In-Plane Devices

The in-plane platform offers distinct advantages for on-chip integration, making it an essential approach for achieving integration of complex functionalities within limited thickness. Its scalable architecture accommodates a wide range of application needs and ensures seamless connectivity with other components, such as light sources and detectors, enabling efficient signal processing in resource-constrained environments. This makes it a promising pathway for developing efficient, and highly controllable devices across various technical fields. Common in-plane structures, such as waveguides, microrings, microdisks, and photonic crystal cavities, will be discussed below.

2.1.1. Waveguides

Nanoscale waveguides serve as fundamental building blocks in integrated optical devices.^[34–37] Like traditional bulk materials,

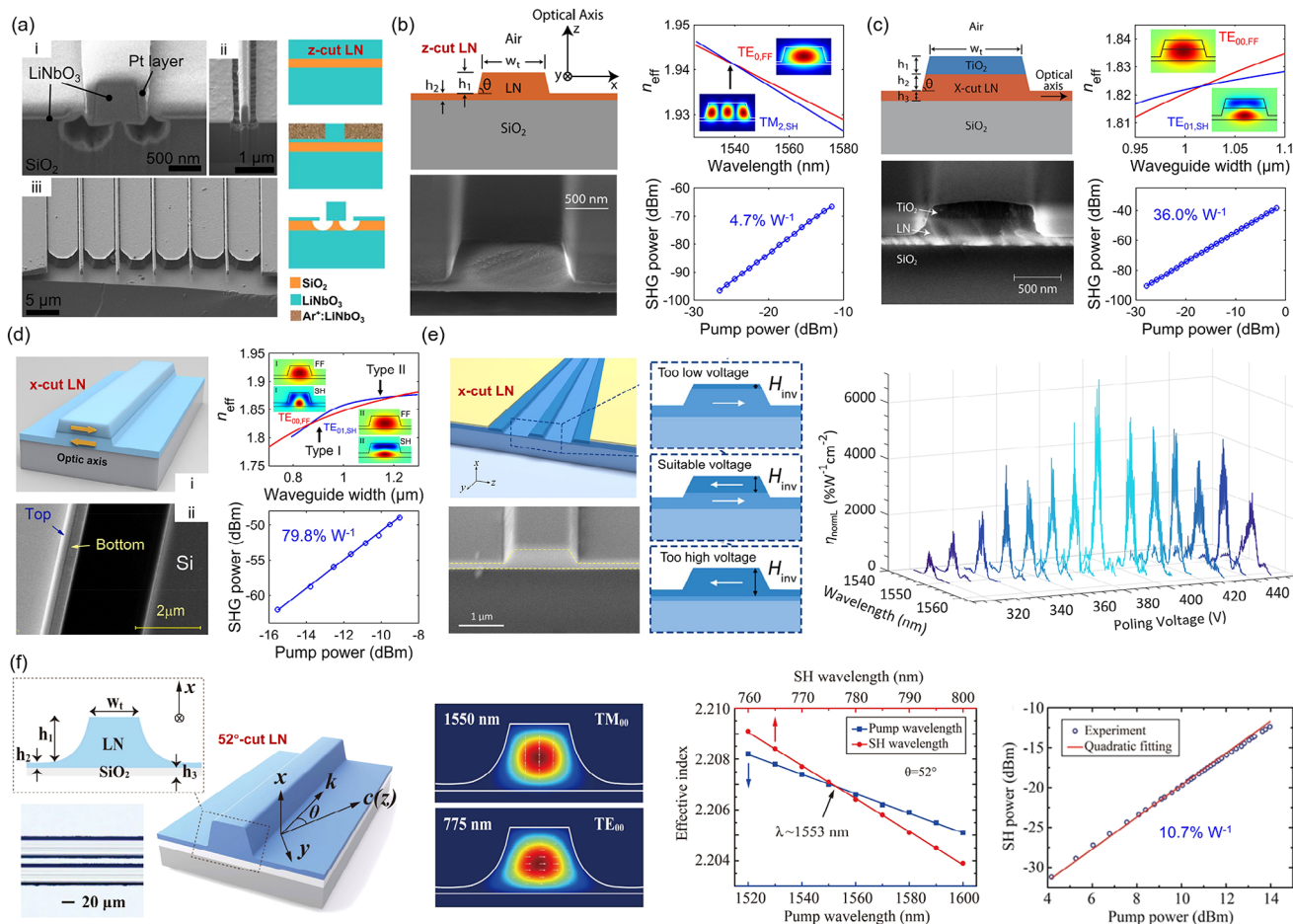


Figure 2. Monolithic LN nanophotonic ridge waveguides for SHG. a) Early implementation of nanophotonic LN waveguide. Reproduced with permission.^[38] Copyright 2015, Optical Society of America. b) Highly tunable, low-loss LN nanophotonic waveguide. Reproduced with permission.^[39] Copyright 2018, Optical Society of America. c) A semi-nonlinear nanophotonic waveguide. Reproduced with permission.^[40] Copyright 2019, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. d) Reverse-polarization dual-layer LN waveguide via wafer bonding. Reproduced with permission.^[41] Copyright 2021, Wiley-VCH GmbH. e) Reverse-polarization dual-layer LN waveguide via in-situ domain engineering. Reproduced with permission.^[42] Copyright 2025, Wiley-VCH GmbH. f) BPM in an angle-cut LN waveguide. Reproduced with permission.^[43] Copyright 2022, Optica Publishing Group.

these waveguides exhibit sufficient propagation distances, typically ranging from millimeters to centimeters. This extended propagation range facilitates the accumulation of nonlinear responses. Compared to their bulk counterparts, wherein the focusing of laser is constrained by the limited Rayleigh range, waveguides offer the capability to tightly confine light fields within micro- to nano-scale areas along their entire propagation path. This confinement significantly enhances light intensity and eliminates the walk-off effect, thereby increasing both the strength and the effective interaction distance of light and matter. As illustrated in **Figure 2a**, millimeter-long nanoscale ridge waveguides with a height of 530 nm and different widths were fabricated using KOH-assisted ion beam-enhanced etching (IBEE) technology on LNOI substrates. Notably, light with a wavelength of about 1.4 μm propagated efficiently in these thin waveguides, highlighting the distinctive advantage of subwavelength field confinement.

Furthermore, at the heart of the efficient SHG is PM condition. The waveguides offer a unique mechanism to achieve phase matching between the FF wave and SH wave, known as modal

phase matching (MPM). This technique leverages the modal dispersion of optical waveguides, which support multiple modes. By adjusting geometric dimensions, the effective refractive index of the SH mode (n_2) can be flexibly made equal to that of the FF mode (n_1), as illustrated by the crossing of effective refractive index curves in **Figure 2b–d**. This MPM technique has been applied to the waveguides in **Figure 2a**. However, these waveguides encountered significant propagation loss due to the exploratory nature of LN processing technology during that period. Reported propagation losses of 61 dBcm⁻¹ for FF and 256 dBcm⁻¹ for SH, attributed to sidewall roughness, severely hampered SHG efficiency, resulting in a normalized efficiency η_{norm} of 0.06% W⁻¹ for a length of 0.9 mm.^[38] This value represents on-chip efficiency, excluding the impact of external coupling losses, such as those arising from fiber-to-chip or objective-to-chip coupling. For clarity, in the following text, the efficiency of in-plane devices will be listed as on-chip efficiency unless otherwise specified. Subsequent years witnessed rapid advancements in nanofabrication technology, leading to substantial improvements in this regard.^[44–47] For instance, LNOI waveguides exhibited high

optical quality with propagation losses reduced to as low as 0.54 dBcm⁻¹ in the telecom band, as depicted in Figure 2b, resulting in a η_{norm} of 4.7% W⁻¹ in an 8-mm-long waveguide. Noteworthy is the exploitation of LN's strong thermo-optic birefringence to achieve thermal tuning of the SHG wavelength, with a tuning slope of 0.84 nmK⁻¹, thereby offering a tunable coherent light source for diverse applications.^[39]

This technique leverages the modal dispersion of optical waveguides, which support multiple modes. Despite the evident advantage of strong mode confinement and flexible MPM, the SHG efficiency of the waveguides is limited by the spatial mode overlap factor between the FF and SH modes. For a lossless waveguide with no pump depletion, the η_{norm} of SHG in the waveguide can be expressed as^[39,41]

$$\eta_{\text{norm}} = \frac{8\pi^2 d_{\text{eff}}^2 L^2 \zeta^2}{c\epsilon_0 n_1^2 n_2 \lambda^2 A_{\text{eff}}} \quad (3)$$

where d_{eff} is the effective nonlinear susceptibility. λ is the FF wavelength. A_{eff} is the effective mode area. The overlap factor ζ represents the spatial overlap between the FF and SH modes, given by:

$$\zeta = \frac{\iint_{\chi^{(2)}} (E_{1\perp}^*)^2 E_{2\perp} dS}{|\iint_{\chi^{(2)}} |E_1|^2 E_1 dS|^{\frac{2}{3}} |\iint_{\chi^{(2)}} |E_2|^2 E_2 dS|^{\frac{1}{3}}} \quad (4)$$

where $E_{1\perp}$ and $E_{2\perp}$ are the transversal components of the electric fields of the FF and SH modes, respectively, while dS denotes the differential area element of integration. Specifically, while the electric field of the SH mode undergoes polarity changes across the cross-section of the waveguide, perpendicular to the propagation direction, the FF mode maintains a consistent polarity. This polarity mismatch results in out-of-phase nonlinear parametric interactions across different regions of the waveguide cross-section, leading to mutual cancellation. Consequently, this significantly reduces the ζ value, thereby minimizing the net nonlinear effect.^[40] To address this inherent limitation, novel semi-nonlinear waveguides comprising a hybrid structure of TiO₂/LN films were proposed.^[40] As illustrated in Figure 2c, these waveguides maintain the same geometry as their monolithic counterparts, but incorporate a linear material layer with an identical refractive index replacing the top portion of the core, where $\chi^{(2)}$ vanishes. Since the linear top layer does not participate in the nonlinear interaction, the parametric process is solely driven by the bottom nonlinear layer, eliminating cancellation effects. This design yields a significantly enhanced net nonlinear effect, with an impressive η_{norm} of 36.0% W⁻¹ in an x -cut LN waveguide of 2.35 mm length, surpassing conventional MPM monolithic waveguides by about one order of magnitude.^[40] Most recently, marvelous double-layer x -cut LN waveguides with reversed polarization were fabricated, wherein the $\chi^{(2)}$ susceptibility within the up/down nodes region is reversed. This innovative design ensures in-phase generation of nonlinear fields in the top and bottom parts of the core, minimizing mutual cancellation between side nodes.^[41,48] Consequently, this approach enabled the realization of a high η_{norm} of 79.8% W⁻¹ in a 1.2 mm-long waveguide.^[41] Notably, this method involves complex preprocessing steps, such as precise wafer alignment, bonding, and wet etching. Addition-

ally, each customized wafer is constrained to a specific MPM design. In contrast, in-situ domain engineering offers a more flexible solution. As shown in Figure 2e, by adjusting the poling voltage, the domain inversion depth can be precisely controlled, significantly simplifying the fabrication process of the double-layer TFLN wafers. As a result, an η_{norm} of up to 1,328% W⁻¹ is achieved in a 4-mm-long waveguide, with an η exceeding 40%.^[42] Another approach to MPM, birefringent modal phase matching (BMPM), utilizes the birefringence of nonlinear crystals. By carefully selecting the crystal orientation and light polarization configuration, the refractive indices of the FF and the SH can be equal, enabling PM. In Figure 2f, a 20-mm-long 52°-cut ridge waveguide achieves an η_{norm} of 10.7% W⁻¹, along with a thermally tunable PM wavelength.^[43] However, the efficiency of such technique is limited by its inability to exploit the largest nonlinear coefficient, d_{33} of LN.

Referring to the strategy of QPM based on bulky periodically poled lithium niobate (PPLN), the idea of PPLN waveguides has also gained considerable attraction and appeal in recent years. QPM relies on the periodic modulation of nonlinear optical coefficients, such as by periodically inverting the ferroelectric domains of the LN. The reciprocal lattice vector of this periodically poled structure compensates for wave vector mismatching in nonlinear wave mixing, thereby effectively achieving PM.^[23] These PPLN waveguides are characterized by sub-wavelength optical confinement and meticulously engineered periodic poling, which exhibit normalized conversion efficiencies surpassing those of previously mentioned devices. The PM condition, written as

$$\frac{\pi}{\Lambda} = \frac{2\pi(n_2 - n_1)}{\lambda} \quad (5)$$

allows for flexible design of the poling period Λ to accommodate desired wavelengths λ . Initially demonstrated on SiN loaded PPLN waveguides before transitioning to monolithic LN waveguides, this method has shown remarkable results.^[49] Figure 3a illustrates the first dry-etched PPLN waveguide, featuring a poling period tailored for first-order QPM to compensate for phase mismatch and facilitate efficient energy transfer over extended interaction lengths.^[49] With a length of 4 mm, the measured η_{norm} reached up to 417% W⁻¹, achieving an absolute conversion efficiency η of 53% under a pump power of 220 mW.^[49] Figure 3b shows PPLN waveguides produced through actively monitored ferroelectric domain reversal engineering, ensuring precise domain engineering via real-time monitoring during manufacturing. However, due to the short length of mere 0.3 mm, these devices achieved up to 4% W⁻¹ efficiency.^[50] Subsequent advancements in PPLN waveguide fabrication have yielded impressive results. Figure 3c depicts a PPLN ridge waveguide with an optimized duty cycle using the same active domain structure monitoring method, achieving an exceptional SHG conversion efficiency of 1,100% W⁻¹ over a 6 mm length.^[51] Notably, the fidelity of poling over extended lengths has been demonstrated using a non-destructive technique of confocal scanning SH microscopy, as shown in Figure 3d.^[52] After edge polishing, the coupling loss was reduced, resulting in high efficiency up to 939% W⁻¹ with a waveguide length of 5 mm.^[52] Recently, micro-waveguides on the 3 μ m thick LNOI platform have been proposed, offering

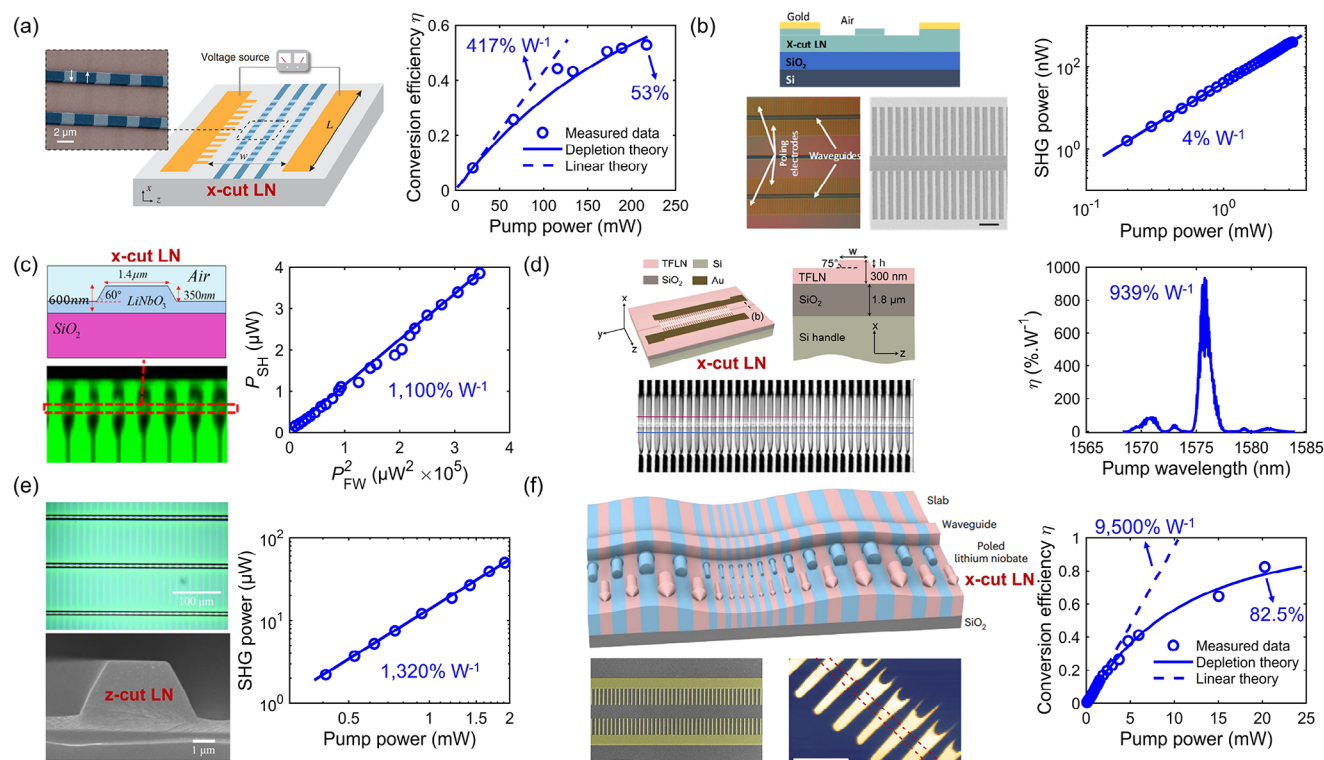


Figure 3. Polled thin film LN waveguides for SHG. a) A dry-etched PPLN waveguide. Reproduced with permission.^[49] Copyright 2018, Optical Society of America. b) An actively-monitored PPLN photonic waveguide. Reproduced with permission.^[50] Copyright 2019, Optical Society of America. c) Efficiency optimization of a PPLN waveguide via in situ monitoring of ferroelectric domains. Reproduced with permission.^[51] Copyright 2020, AIP Publishing. d) Shallow-etched PPLN waveguides. Reproduced with permission.^[52] Copyright 2020, Optical Society of America. e) Fiber-compatible PPLN micro-waveguides. Reproduced with permission.^[53] Copyright 2023, Optica Publishing Group. f) Adaptively poled LN waveguides. Reproduced with permission.^[54] Copyright 2024, CC BY 4.0, under exclusive licence to Springer Nature Limited.

a balanced combination of high performance, easy machining, and efficient fiber coupling, as shown in Figure 3e.^[53] With a fiber-waveguide coupling loss of approximately 1 dB/facet, these waveguides exhibited an efficiency exceeding $1,320\% W^{-1}$ over a 25 mm length.^[53] However, challenges persist with inhomogeneous thickness across the LNOI wafer, which limits the effective coherent interaction length in PPLN waveguides.^[54] To address this, an adapted poling approach has been developed, as illustrated in Figure 3f. This approach mitigates wafer inhomogeneity, achieving a peak efficiency of $9,500\% W^{-1}$ for 21-mm-long waveguides, closely approaching the theoretical limit of $10,500\% W^{-1}$. Additionally, an absolute efficiency exceeding 80% was demonstrated in a single-pass configuration with a pump power of 20 mW.^[54]

In addition to the approaches discussed above, several other innovative strategies have been proposed to achieve PM in waveguides. For instance, Figure 4a illustrates the concept of periodically grooved LN (PGLN) waveguides.^[55] By introducing periodic modulation of the waveguide width, an additional momentum, proportional to the reciprocal of the period, compensates for phase mismatch. An η_{norm} of $0.017\% W^{-1}$ was achieved. Furthermore, Figure 4b presents a hybrid nonlinear integrated photonic device consisting of phase gradient metasurfaces patterned on top of a nonlinear waveguide. This device capitalizes on the capability of gradient metasurfaces to control modal indices and

the extensive light-matter interaction provided by optical waveguides. The unidirectional effective wavevector provided by gradient metasurfaces enables a one-way transfer of optical power from the pump to the SH signal. The measured η_{norm} achieved $6.0 \times 10^{-5} W^{-1}$ under a short metasurface-covered device length of only 19 μ m.^[56] The $\eta_{\text{norm,L}}$ achieved $1,660\% W^{-1} \text{cm}^{-2}$, significantly higher than conventional PPLN and recent reports on thin film LN waveguides due to the strong nonlinear modal overlap in their devices.

In contrast to ridge LN waveguides, which encounter challenges in achieving high-quality lithographic patterning and dry etching of LN, rib-loaded LN waveguides have emerged as a viable alternative. These waveguides circumvent the need for LN etching by incorporating a strip of material that is deposited or patterned on top of the unetched LN thin film, as illustrated in Figure 5. This configuration forms an effective rib waveguide, where part of the optical mode is confined and guided within the LN layer.^[57] As depicted in Figure 5a, a polymer waveguide with carefully chosen dimensions was patterned on an LNOI substrate. This approach is based on the concept of bound states in the continuum (BIC), where waveguiding is achieved by destructive interference of all leakage channels. These waveguides facilitate MPM between orthogonally polarized FF and SH modes, both of which exhibit low propagation losses. In a 5-mm-long waveguide, a measured maximal η_{norm} of $0.044\% W^{-1}$ was

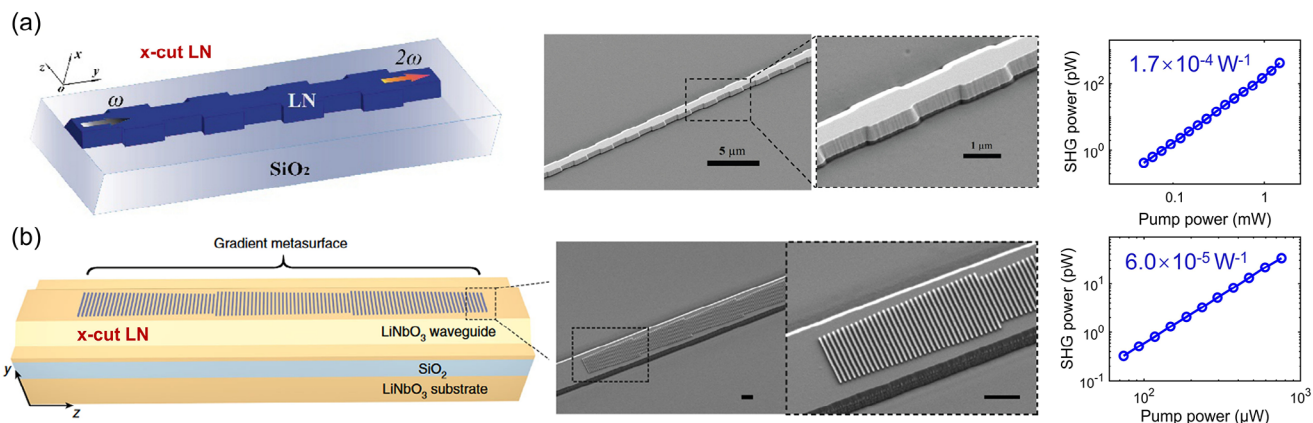


Figure 4. Design-assisted LN waveguides for SHG. a) QPM in a periodically grooved LN waveguide. Reproduced with permission.^[55] Copyright 2017, Optical Society of America. b) Metasurface-assisted LN waveguide for phase-matching-free SHG. Reproduced with permission.^[56] Copyright 2017, CC BY 4.0.

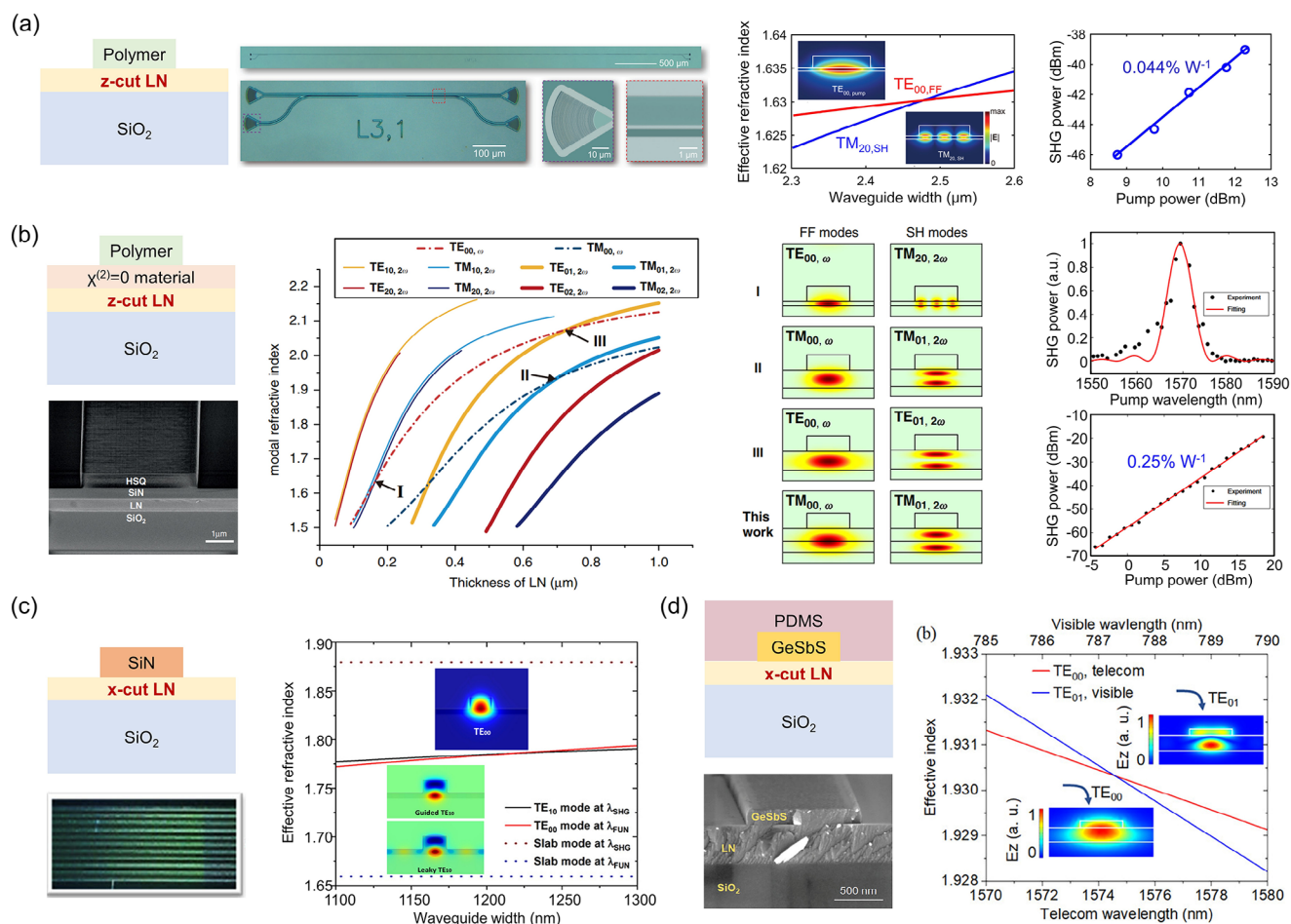


Figure 5. Rib-loaded LN nanophotonic waveguides for SHG. a) Polymer rib-loaded LN nanophotonic waveguides. Reproduced with permission.^[58] Copyright 2022, Wiley-VCH GmbH. b) HSQ rib-loaded LN waveguides. Reproduced with permission.^[59] Copyright 2022, CC BY 4.0. c) SiN rib-loaded LN waveguide. Reproduced with permission.^[60] 2024, CC BY 4.0. Laser & Photonics Reviews published by Wiley-VCH GmbH. d) GhG rib-loaded LN waveguide. Reproduced with permission.^[61] 2024, Wiley-VCH GmbH.

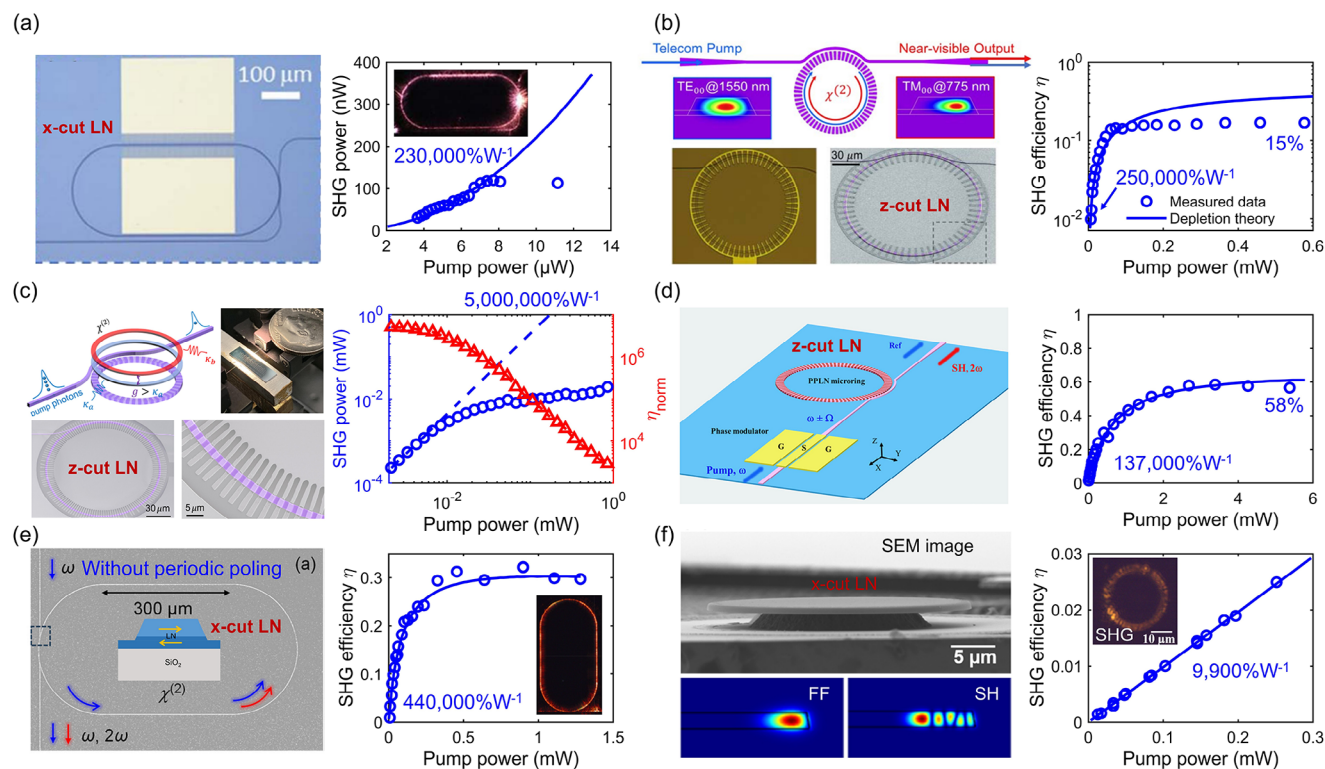


Figure 6. LNOI Microrings and microdisks for SHG. a) A racetrack shape with one PPLN arm straight. Reproduced with permission.^[63] Copyright 2019, Optical Society of America. b) PPLN microring resonators. Reproduced with permission.^[64] Copyright 2019, Optical Society of America. c) Optimized PPLN microring resonators. Reproduced with permission.^[65] Copyright 2020, Optical Society of America. d) Efficient on-chip frequency doubling with active stabilization. Reproduced with permission.^[66] Copyright 2021, Wiley-VCH GmbH. e) An LN microcavity without periodic poling. Reproduced with permission.^[67] Copyright 2024, Wiley-VCH GmbH. f) On-chip monocrystalline LN microdisk resonator. Reproduced with permission.^[69] Copyright 2019, American Physical Society.

attained.^[58] Similarly, Figure 5b shows etchless semi-nonlinear silicon nitride SiN/LN photonic waveguides, which offer optimized nonlinear modal overlaps and η_{norm} of up to 0.25% W⁻¹ in a 2.5-mm-long waveguide.^[59] Further studies, as shown in Figure 5c, have demonstrated that SiN-loaded x-cut LN waveguides not only enable strong mode overlap between the vertically antisymmetric TE₁₀ SH mode and the vertically symmetric TE₀₀ FF mode, but also effectively utilize the BIC mechanism to suppress SH mode leakage. By optimizing the waveguide geometry for precise phase matching, a normalized efficiency η_{norm} reached 3% W⁻¹.^[60] Additionally, as shown in Figure 5d, a strip of chalcogenide (ChG) material was deposited onto an unetched x-cut LN thin film, significantly enhancing SHG efficiency. An experimental overall conversion efficiency η_{norm} of 25.55% W⁻¹ was achieved in a 1-cm-long waveguide.^[61]

2.1.2. Microrings and Microdisks

By bending waveguides into closed loops, microring cavities are constructed. These cavities inherit the spatial light confinement benefits of waveguides, operating with micro- to nano-scale cross-sectional areas.^[62] Furthermore, the light can traverse multiple passes through the cavity, extending the effective propagation distance of light. Light-material interactions are substan-

tially enhanced compared to open-looped waveguides. **Figure 6a** shows a racetrack-shaped structure with one arm consisting of a straight PPLN waveguide. This configuration achieves double resonances at both FF and SH wavelengths, meeting stringent requirements for ultra-efficient nonlinearities from cavity. An absolute efficiency of 1.3% was reached with only 5.6 μW of pump power, corresponding to a η_{norm} of 230,000% W⁻¹.^[63] Furthermore, the PPLN microring resonator shown in **Figure 6b** demonstrated an on-chip SHG efficiency of up to 250,000% W⁻¹ before saturation.^[64] Due to the experimental feasibility of a large poling period and the potential high Q factor of the pump TE₀₀ mode, this work initially focused on the FF-TE₀₀ to SH-TM₀₀ conversion using the d_{31} tensor component. By optimizing the cavity design, as shown in **Figure 6c**, TM₀₀-to-TM₀₀ SHG was realized, accessing the largest d_{33} tensor component. This optimization led to a 20-fold increase in efficiency, reaching 5,000,000% W⁻¹ in a low power regime of below 10 μW. This remains the record-high efficiency to date among current integrated photonic platforms.^[65] Furthermore, through the optimal integration of a z-cut electro-optic phase modulator and a PPLN microring on the same chip, as shown in **Figure 6d**, stable SHG was achieved at approximately 60% conversion efficiency using milliwatt pump power, employing simultaneous overcoupling for both FF and SH modes. This device addresses long-standing challenges in cavity-based optical processing, such as frequency conversion, frequency comb

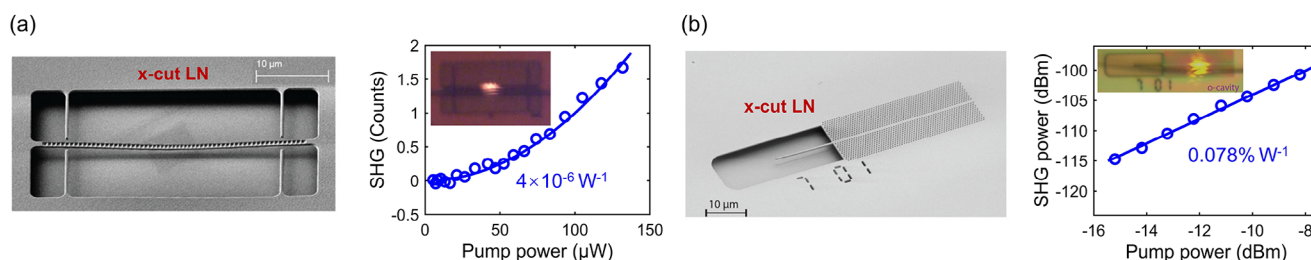


Figure 7. On-chip LN photonic crystal nanocavities for SHG. a) 1D LN photonic crystal nanocavities. Reproduced with permission.^[75] Copyright 2018, AIP Publishing. b) High-Q 2D LN photonic crystal nanocavities. Reproduced with permission.^[76] Copyright 2019, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

generation, and all-optical switching, which often suffer from stability issues due to photorefractive or thermal effects. These issues are effectively mitigated through electro-optic modulation in this demonstration.^[66] Additionally, as shown in Figure 6e, double-layer microrings with reversed polarization have been employed to achieve high SHG efficiency without need for poling, further enhancing the potential of these systems in practical applications.^[67,68]

High-Q factor microdisks are also promising candidates for efficient nonlinear frequency conversion due to their ability to support unconventional whispering-gallery modes, distinct from those in microrings.^[70,71] These devices benefit from smooth boundary fabrication, which minimizes optical losses and enhances their performance. However, achieving efficient wavelength conversion in microdisks presents a significant challenge due to the limited control over dispersion, which is crucial for satisfying the stringent PM condition necessary for efficient nonlinear processes.^[72] A smart solution to this challenge involves a unique broadband natural QPM mechanism, as depicted in Figure 6f. For TE mode light waves propagating along the circumference of the microdisk, the effective nonlinear optical coefficients naturally oscillate periodically, changing both in sign and magnitude. This intrinsic periodic oscillation facilitates QPM without requiring domain engineering in the micrometer-scale LN disk. Consequently, the SH waves were generated with efficiencies as high as $9,900\% \text{ W}^{-1}$, with the SH beam visibly encircling the 30 μm diameter of the LN microdisk.^[69] This approach to QPM leverages the inherent properties of the microdisk, providing a robust and efficient method for nonlinear frequency conversion. The ability to generate SH waves efficiently without complex domain engineering underscores the potential of microdisks in advanced photonic applications, where precise control over light-matter interactions is paramount.

2.1.3. Photonic Crystal Cavities

Photonic crystal nanocavities exhibit superior light confinement within subwavelength dimensions, making them promising for nonlinear photonic applications.^[73,74] Figure 7a, b illustrate the 1D and 2D on-chip LN photonic crystal nanoresonators, respectively. The FF light was launched into the photonic crystal nanocavities via tapered optical fibers which also deliver the generated SH light output from the devices. These structures com-

bine high Q-factors with tight optical mode confinement, enabling significant nonlinear optical interactions and facilitating SHG at excitation power levels as low as tens of microwatts.^[75,76] The reported fiber-to-fiber conversion efficiencies, which account for coupling losses between the fibers and the nanocavity, remain below $0.08\% \text{ W}^{-1}$. This low efficiency has been attributed to limitations in both the coupling process and the photonic crystal design. Specifically, the tapered optical fiber, originally optimized for FF coupling, exhibits poor coupling efficiency at the SH wavelength, leading to significant transmission losses. Additionally, while the photonic crystal structures were designed to support high-Q cavities at the FF; their Q-factors at the SH wavelength are substantially lower. This reduction occurs because the SH mode lies above the light line, resulting in considerable radiation leakage into free space. To enhance nonlinear conversion efficiency, future research should focus on optimizing both the external coupling and the photonic crystal resonance of the SH wave. These improvements are crucial for achieving higher collection efficiency and boosting the overall SHG performance in these systems.

2.1.4. Section Summary

As a short summary, leveraging the small mode volume and various PM technologies has enabled significant advancements in achieving high-efficiency SHG with low power consumption on-chip. These devices demonstrate superior performance for frequency conversion of continuous wave lasers and open new possibilities for nonlinear science at the nanoscale and weak pump power level. To provide a comprehensive comparison, we have summarized the main performance parameters of these in-plane devices in Table 1.

2.2. Out-of-Plane Devices

While significant advancements have been made in achieving high-performance SHG through in-plane devices as introduced above, there exists a compelling need for out-of-plane SHG devices. In these configurations, SHG waves are generated and manipulated in the direction normal to the wafer plane. Such devices hold promise for applications in displays, optical data storage, free-space optical communications, and so on. This out-of-plane approach effectively compresses the light-matter interaction distance to the subwavelength scale, resulting in a

Table 1. LNOI based in-plane SHG devices.

Structure and reference	Length [mm]	Fiber-to-chip coupling loss for pump/SHG [dB/facet]	η_{norm} [%W ⁻¹]	$\eta_{\text{norm,L}}$ [%W ⁻¹ cm ⁻²]	η
Ridge waveguide ^[38]	0.9	-/-	0.06	7.4	4.1×10^{-7}
Ridge waveguide ^[39]	8	5/-	4.7	7.3	3.3×10^{-6}
TiO ₂ /LN dual-layer waveguide ^[40]	2.35	5.9/-	36.0	650	2.3×10^{-4}
Reverse-polarization dual-layer waveguide ^[41]	1.2	7.4/-	79.8	5540	1.0×10^{-4}
Reverse-polarization dual-layer waveguide ^[42]	4	4.7/12.5	1328	8300	40.5%
MBMP waveguide ^[43]	20	-/-	10.8	2.7	2.4×10^{-3}
Periodically grooved waveguide ^[55]	0.5	4.8/-	0.017	6.8	7.5×10^{-9}
Metasurface-assisted waveguide ^[56]	0.019	-/-	0.006	1660	4.3×10^{-8}
Polymer rib-loaded waveguide ^[58]	5	9.3/18.3	0.044	0.175	7.3×10^{-6}
HSQ rib-loaded SiN/LN waveguide ^[59]	2.5	11.45/9.65	0.25	4.05	1.8×10^{-4}
SiN rib-loaded waveguide ^[60]	1	5.5/6	3	300	1.2×10^{-4}
ChG rib-loaded waveguide ^[61]	10	4.62/-	25.55	25.55	7.2×10^{-4}
SiN rib-loaded PPLN waveguide ^[77]	5	6/14	40	160	1.6×10^{-4}
PPLN ridge waveguide ^[49]	4	10/10	417	2,600	53%
PPLN ridge waveguide ^[50]	0.3	6/7	4	4,600	1.3×10^{-4}
PPLN ridge waveguide ^[51]	6	10/1	1,100	3,061	6.6×10^{-3}
Edge polishing PPLN waveguide ^[52]	5	6.7/12.5	939	3,757	9.4%
PPLN micro-waveguide ^[53]	23	1/-	1,320	219	2.6%
Adaptively poled waveguides ^[54]	21	10.7/14	9,500	2,150	82.5%
A racetrack shape PPLN microring ^[63]	—	5.5/8.5	230,000	—	6%
PPLN microring ^[64]	*0.07	-/-	250,000	—	15%
Optimized PPLN microring ^[65]	*0.07	8.5/10	5,000,000	—	5.3%
Actively stabilized PPLN microring ^[66]	*0.08	17/16	137,000	—	58%
Reverse-polarization dual-layer microring ^[67]	—	8.4/8.0	440,000	—	30%
Reverse-polarization dual-layer microring ^[68]	—	17.8/-	149,000	—	10.3%
Microdisk ^[69]	*0.015	-/-	9,900	—	3.7%

Notes: Efficiency values represent on-chip efficiency, excluding the impact of external coupling losses, such as those arising from fiber-to-chip coupling. *The radius of the microring or microdisk.

ultra-compact design. Due to their subwavelength thicknesses, these out-of-plane devices do not require PM considerations; however, this same characteristic also renders traditional PM strategies largely ineffective, which rely on a macroscopic propagation length to enhance efficiency. Instead, in such out-of-plane devices, the enhancement of nonlinear efficiency is primarily driven by the strong localization of the electric field rather than by cumulative propagation effects.^[78] In this context, the enhancement of nonlinear efficiency is achieved primarily through the strong localization of the electric field, denoted as E , at the resonant frequencies of the device. This localization is critical for maximizing the interaction strength between the light and the material, ultimately facilitating enhanced nonlinear optical processes. Resonance plays a key role in achieving this localization. Specifically, resonant nanostructures leveraging resonance at the FF wavelength can efficiently couple far-field radiation into subwavelength dimensions, thereby concentrating the incident electromagnetic field and significantly amplifying nonlinear interactions. Meanwhile, when the generated SH signal aligns with the resonance at the SH wavelength, it experiences enhanced nonlinear wave radiation into the far field.

2.2.1. Metasurfaces

Metasurfaces are artificially engineered ultrathin structured interfaces that demonstrate significant potential across a range of applications, including imaging, information processing, and telecommunications. Their ultra-compact nature allows for innovative manipulation of electromagnetic fields, resulting in the formation of resonances at specific wavelengths. These resonances enable nonlinear light-matter interactions, which can be spectrally enhanced with precision.^[79–86]

As illustrated in Figure 8a, b, Mie-type resonances have been successfully achieved in truncated pyramid arrays and nano-grating metasurfaces, respectively, leading to resonantly enhanced SHG. Notably, conversion efficiencies on the order of 10^{-7} to 10^{-6} have been observed under femtosecond laser excitation at intensities of several GWcm⁻².^[87,88] Additionally, by increasing the refractive index contrast between the metasurface layer and the surrounding medium, substantial improvements in electromagnetic field confinement can be realized. As shown in Figure 8c, building on this principle, freestanding LN membrane metasurfaces were fabricated following substrate removal, demonstrating over two orders of magnitude greater

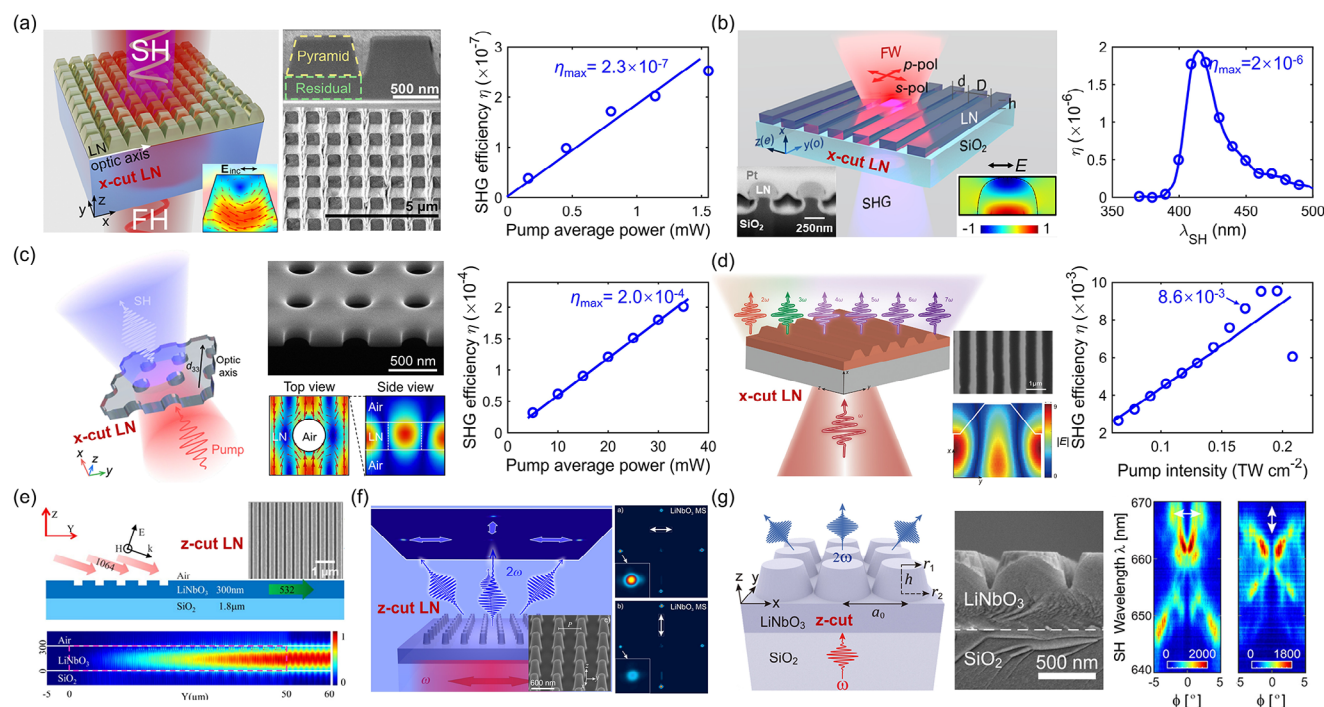


Figure 8. Monolithic LN metasurfaces for SHG. a) A resonant 2D truncated pyramid metasurface. Reproduced with permission.^[87] Copyright 2020, American Chemical Society. b) 1D grating metasurface. Reproduced with permission.^[88] Copyright 2021, Wiley-VCH GmbH. c) Membrane metasurfaces. Reproduced with permission.^[89] Copyright 2022, American Chemical Society. d) Shallow-etched grating metasurface. Reproduced with permission.^[90] Copyright 2023, The Royal Society of Chemistry. e) Slab waveguides coupled by grating metasurfaces. Reproduced with permission.^[91] Copyright 2020, Chinese Laser Press. f) SHG beam steering and polarization encoding by nanopillar array metasurfaces. Reproduced with permission.^[92] Copyright 2021, CC BY 4.0. g) Strong nonlinear enhancement through strong out-of-plane electric field component. Reproduced with permission.^[93] Copyright 2022, Wiley-VCH GmbH.

enhancement compared to previously reported LN metasurfaces, achieving an absolute conversion efficiency of 2×10^{-4} under a pump intensity of 2.4 GW cm^{-2} .^[89] Figure 8d highlights efficient SHG at elevated pump intensities of 170 GW cm^{-2} , yielding a peak conversion efficiency of up to 0.86%, along with the observation of seventh harmonic generation in the deep ultraviolet region.^[90]

One significant advantage of metasurfaces lies in their ability to provide flexible control over the characteristics of SHG, including its spatial distribution, emission direction, and polarization properties. This control is achieved through the precise structuring and arrangement of nanosized meta-atoms. As illustrated in Figure 8e, SHG can be effectively generated from a grating metasurface-assisted slab waveguide, facilitating the conversion from free-space FF wave to guided SH waves.^[91] Further advancements have been made by integrating holographic designs into the metasurface on the slab waveguide, enabling innovative beam shaping techniques such as dual focusing and Airy beam generation. These developments pave the way for multichannel nonlinear conversion and generation, enhancing optical interconnections. Moreover, as shown in Figure 8f, the MD (magnetic dipole) resonance exhibited a strong out-of-plane electric field component, thus maximizing access to the nonlinear d_{33} component along the out-of-plane direction of z-cut LN. This configuration allows for the encoding of the pump light's polarization into specific SH polarization states and diffraction orders.^[92] Similarly, Figure 8g demonstrates that the transverse magnetic (TM)

mode quasi-BIC also possess strong out-of-plane electric field components. By modulating the lattice symmetries, researchers have gained additional degrees of freedom to control SHG, achieving emission of SH light with polarization orthogonal to that of the pump.^[93] In summary, leveraging the unique properties of resonant metasurfaces not only enhances the efficiency of SHG but also provides new avenues for manipulating light at the nanoscale, presenting significant potential for future photonic applications.

In parallel with the previously discussed etchless waveguide systems, the concept of etchless LN metasurfaces has also emerged. This approach involves the realization of periodic structures composed of low-refractive index materials on LN thin films, aiming to bypass the complexities associated with LN etching processes while achieving highly nonlocal resonances characterized by high Q-factors. As depicted in Figure 9a, a silica grating was fabricated on LN thin films with subwavelength thickness, leading to a remarkable 50-fold increase in SHG efficiency compared to an unpatterned film with similar thickness.^[94] Enhanced SHG facilitated by GMR and quasi-BICs is further demonstrated in the configurations presented in Figure 9b,c, respectively.^[95,96] However, it is important to note that, in contrast to fully etched LN metasurfaces, etchless LN structures exhibit limited control over the local mode fields that overlap with nonlinear materials, which may result in comparatively lower efficiencies. Additionally, Figure 9d depicts a dielectric nonlocal metasurface that effectively maintains high Q resonances across a broad spectral

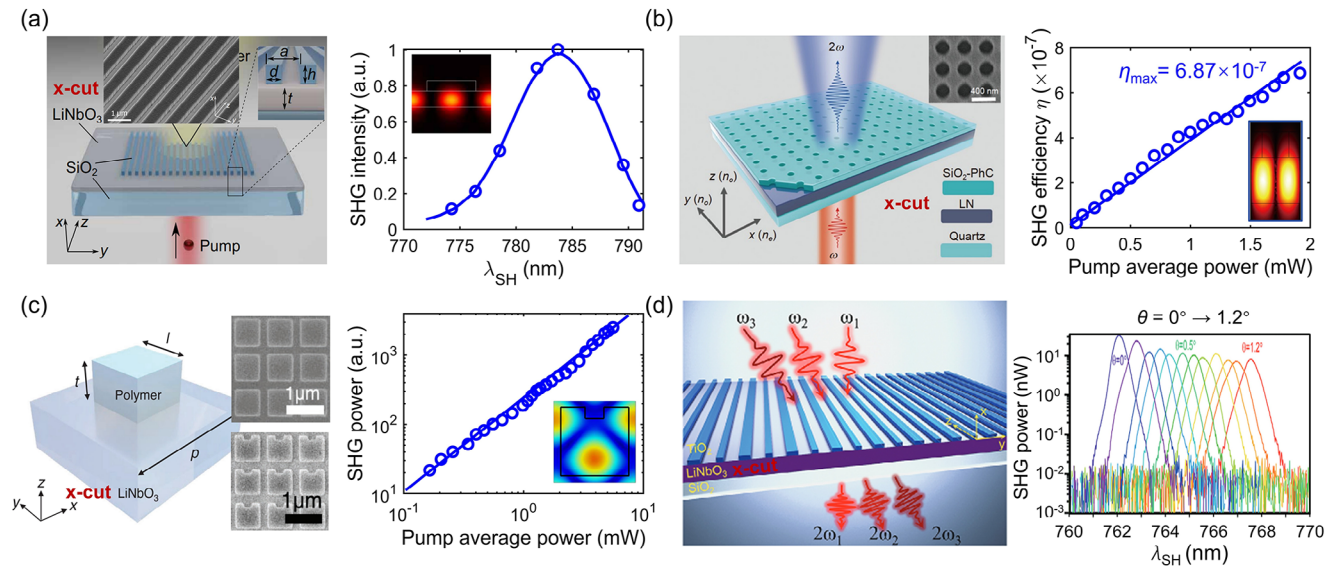


Figure 9. Hybrid etchless LN metasurfaces for SHG. a) A metasurface incorporating a nonlinear thin film LN covered by a silica meta-grating. Reproduced with permission.^[94] Copyright 2022, CC BY 4.0. b) Etchless thin film LN heteronanostructure. Reproduced with permission.^[95] Copyright 2022, Science China Press and Springer-Verlag GmbH Germany. c) A polymer-LN integrated metasurface. Reproduced with permission.^[96] Copyright 2024, Wiley-VCH GmbH. d) Tunable SHG in a nonlocal metasurface over a broad spectral range. Reproduced with permission.^[97] Copyright 2024, Wiley-VCH GmbH.

range, allowing specific wavelengths to be selected by adjusting the incidence angle, thereby enabling spectrally tunable SHG.^[97]

2.2.2. Photonic Crystal Defect Cavities

Photonic crystals represent a distinct class of artificial structures characterized by spatially modulated dielectric functions. By intentionally breaking lattice periodicity, photonic defect states, and corresponding defect cavities can be formed.^[98–100] The introduction of these defect states facilitates strong photon localization, resulting in pronounced hotspots within microcavities that enhance nonlinear light-matter interactions.^[101] As illustrated in

Figure 10a, SHG in a photonic crystal L3 cavity, drilled in suspended LN films, highlights the significant local field enhancement at the defect site, which leads to a substantial increase in nonlinearity at resonance.^[102] Furthermore, Figure 10b demonstrates the integration of guided-mode resonance (GMR) cavities with distributed Bragg reflectors (DBR), resulting in a heterostructure cavity that facilitates efficient coupling and localization of electromagnetic fields. This configuration has achieved a Q-factor exceeding 5000 when subjected to normal incidence of a focused Gaussian beam. Remarkably, an enhancement factor greater than 1200 for SHG has been reported in comparison to unstructured thin LN film under a pump intensity of 1.92 MW cm^{-2} .^[19] In the latest advancement, illustrated in Figure 10c, they

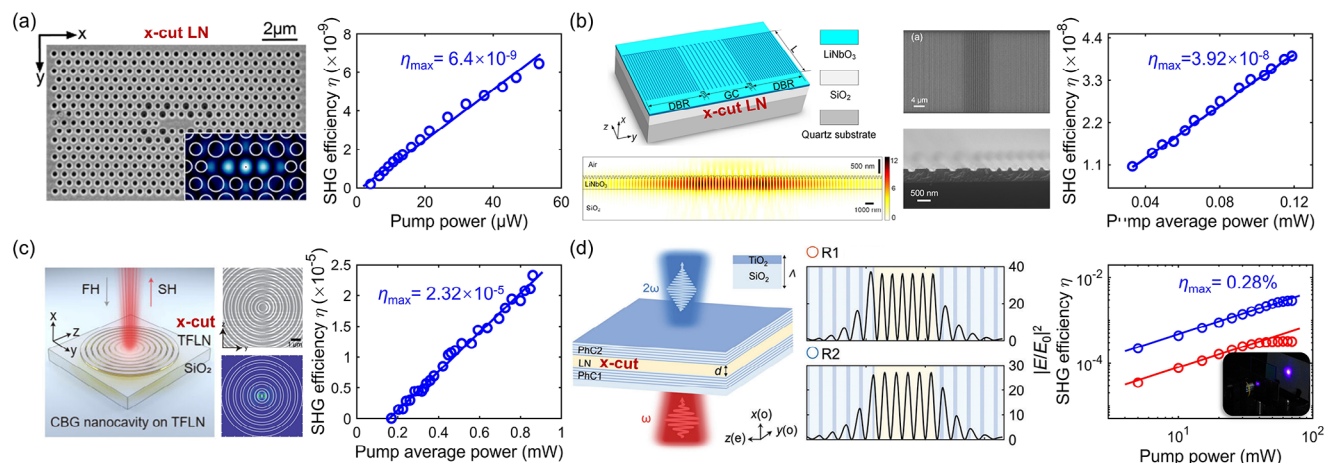


Figure 10. Photonic crystal defect cavities for SHG. a) Photonic crystal L3 cavity. Reproduced with permission.^[102] Copyright 2013, AIP Publishing. b) A heterostructure cavity integrating grating with DBRs. Reproduced with permission.^[19] Copyright 2021, American Physical Society. c) Circular Bragg gratings (CBG). Reproduced with permission.^[103] Copyright 2024, American Chemical Society. d) 1D photonic crystal vertical cavity. Reproduced with permission.^[104] Copyright 2023, Wiley-VCH GmbH.

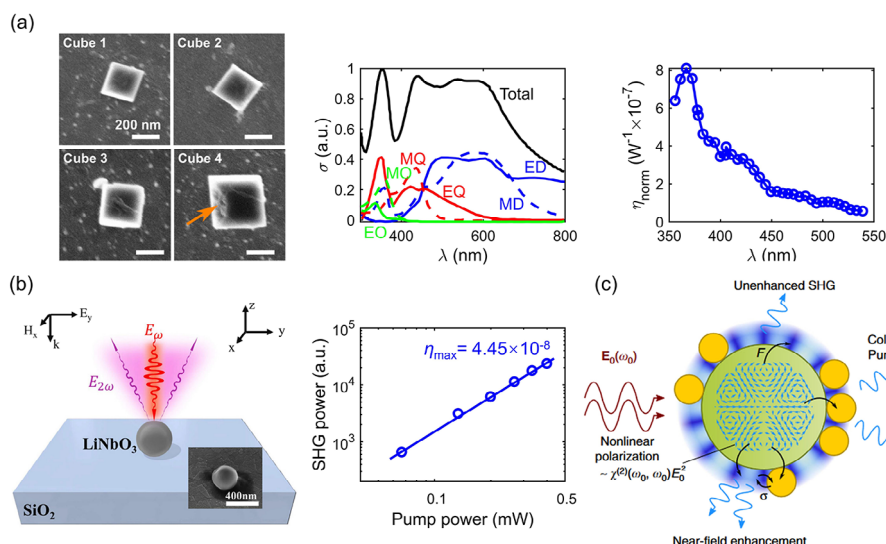


Figure 11. Mie-resonant LN harmonic nanoparticles for SHG. a) LN nanocubes as nonlinear Mie resonators. Reproduced with permission.^[106] Copyright 2019, American Chemical Society. b) Mie-resonant LN nanospheres. Reproduced with permission.^[107] Copyright 2021, CC BY 4.0. c) Near-field enhancement of optical SHG in hybrid gold-LN nanostructures. Reproduced with permission.^[108] Copyright 2023, CC BY 4.0.

achieved a 42,000-fold enhancement in SHG at low pump intensities of 1.9 MW cm^{-2} . This significant improvement is attributed to the small model volume and high light collection efficiency of circular Bragg gratings (CBG), specifically in the carefully designed nanocavities formed by second-order CBG resonators that support anapole modes.^[103] The results above illustrate the design of defect photonic crystals within thin films, effectively confining the electric field within a small volume. In comparison, by stacking alternating layers of high and low refractive index materials, 1D photonic crystals can be constructed, offering simplicity in design and ease of fabrication. When a material slab is embedded as a defect layer into the photonic crystals, the disruption of perfect periodicity generates resonant defect states within the photonic bandgap. As demonstrated in Figure 10d, bright SHG emission was produced from a vertical cavity formed by inserting an LN film between two dielectric photonic crystals. This ingenious design of the cavity, resonating at the FF frequency, resulted in a high SHG efficiency of 0.28%, which is two orders of magnitude greater than previously reported for similar microcavities.^[104] By further integrating a composite grating waveguide structure into a DBR platform that leverages the unique properties of bound states in the Bloch surface wave-enhanced continuum, high-Q will be achieved across the spectral range. This approach has the potential to further enhance SHG efficiency through near-field amplification of the light field, which is tightly localized within the nonlinear material.^[105] Overall, photonic crystal structures that leverage defect-induced resonances represent a promising approach for enhancing nonlinear optical processes, thereby showing their potential in advancing photonic technologies.

2.2.3. Harmonic Nanoparticles

Harmonic nanoparticles, including nanocubes and nanospheres, are nanoscale materials that exhibit nonlinear optical properties,

enabling the generation of SH light. These nanoparticles, emitting coherent light, are used in various applications, such as probes for multiphoton imaging and markers in advanced microscopy. Their advantages over traditional fluorescent probes include reduced photobleaching, diminished nonradiative losses, and superior spectral separation between the emitted signal and the incident light. Their unique optical characteristics significantly enhance imaging resolution and sensitivity, particularly in complex biological environments. Moreover, metal oxide nanoparticles are robustness and chemically inert, which contribute to their biocompatibility and nontoxicity.

Nanoparticles possess the capability to locally confine electromagnetic fields, resulting in complex linear extinction spectra characterized by various electric and magnetic multipole resonances.^[18] These resonances can be harnessed to enhance light-matter interactions, thereby improving the efficiency of nonlinear signal emission. For instance, Figure 11a illustrates LN nanocubes fabricated by solvothermal synthesis. The presence of multiple Mie resonances within these nanocubes significantly contributes to the overall scattering cross-section, with their tunability intricately linked to the dimensions of the nanocube. Typically, the linear response of Mie resonances is analyzed using multipole expansion. The total scattering curve corresponds to the transmission spectrum, while the expanded results reveal the contributions of various multipoles at different resonance peaks. For instance, the linear scattering cross-section at 550 nm results from the overlapping contributions of electric dipoles (ED), magnetic dipoles (MD), and magnetic quadrupoles (MQ). Meanwhile, the peak observed at 350 nm is attributed to the combined effects of magnetic dipoles (MD), electric quadrupoles (EQ), electric octupoles (EO), and magnetic octupoles (MO). Notably, high-order Mie resonances can substantially enhance SHG emission at ultraviolet (UV) spectral range below 400 nm. The SHG from single LN nanocubes was reported to be enhanced by up to 10^7 times compared to bulk LN at 360 nm. Further polarimetric characterization revealed that LN nanocubes exhibit predictable SHG

Table 2. LNOI based out-of-plane SHG devices.

Structure and reference	R [MHz]	τ [fs]	P_1 [mW]	I_{pk} [GWcm ⁻²]	η	η_{norm} [W ⁻¹]	$\eta_{norm,l}$ [cm ² /GW]
Truncated pyramids ^[87]	0.5	540	0.88	4.3	2.3×10^{-7}	5.2×10^{-11}	5.3×10^{-8}
Nanograting ^[88]	80	200	23	2.05	2.0×10^{-6}	1.0×10^{-9}	9.8×10^{-7}
Membrane metasurface ^[89]	80	230	35	2.4	2.0×10^{-4}	8.0×10^{-8}	8.3×10^{-5}
Shallow-etched grating ^[90]	0.075	290	37.6	170	8.6×10^{-3}	3.7×10^{-9}	5.1×10^{-5}
Grating slabs ^[91]	40	200	25	—	4.6×10^{-7}	—	—
Nanocylinder ^[92]	80	140	10	0.5	2.5×10^{-8}	3.0×10^{-11}	5.0×10^{-8}
SiO ₂ /LN heteronanostructure ^[95]	80	140	0.045	2	6.9×10^{-7}	2.9×10^{-9}	1.5×10^{-5}
Grating and DBR heterostructure ^[19]	100	90	0.12	0.0019	3.9×10^{-8}	2.2×10^{-9}	1.3×10^{-6}
Circular Bragg grating ^[103]	20	2.2×10^5	0.86	0.0019	2.3×10^{-5}	9.0×10^{-5}	1.2×10^{-2}
1D vertical photonic crystal ^[104]	80	230	60	4.2	2.8×10^{-3}	6.4×10^{-7}	6.7×10^{-4}
Nanospheres ^[107]	76	130	0.4	2.0	4.5×10^{-8}	8.3×10^{-10}	2.3×10^{-8}

Note: R - Laser repetition rate, τ - Laser pulse duration

emission patterns and crystal orientations when deposited with a face parallel to a flat substrate. This orientation-dependent behavior facilitates better control over the design of nanostructures for future applications.^[106]

Similarly, Figure 11b illustrates the UV SHG from single LN nanospheres fabricated by the fs laser ablation technique. In these studies, individual LN nanospheres function as Mie resonators, facilitating the development of simple yet efficient nonlinear nanophotonic devices. By exploiting the resonant properties of these nanoparticles, significant enhancements in nonlinear optical processes can be achieved. Supporting the concept of Mie-resonance, single LN nanospheres exhibit a field enhancement of 5.5, resulting in a SHG conversion efficiency measured as 4.45×10^{-8} under excitation of a fs laser operating at 750 nm. The tunability and strong field localization provided by these nanospheres further present their potential in the development of high-performance, miniaturized photonic devices.^[107] Furthermore, the SHG from individual LN particles can be significantly enhanced through nonradiative coupling to the localized surface plasmon resonances of embedded gold nanoparticles, as shown in Figure 11c.^[108] Additionally, optical-trapped LN nanoparticles can generate SHG using a substrate-free, liquid-free scanning method for rapid assessment in natural solvents. Liquid solutions flow through microfluidic channels before laser capture, independent of the SHG source, making it suitable for active substrateless metasurfaces.^[109]

2.2.4. Section Summary

Table 2 summarizes the SHG performance of LN out-of-plane devices. Out-of-plane nonlinear devices utilize artificially engineered structures with subwavelength thickness, distinguishing them from in-plane designs that rely on long-distance propagation within a planar geometry. This ultracompact thickness significantly shortens the propagation length, thereby relaxing strict phase-matching requirements and enabling greater design flexibility for tailoring nonlinear interactions through precise structural engineering. Specifically, metasurfaces leverage their structural tunability to manipulate the wavefront of nonlinear optical waves, facilitating versatile beam shaping and high-efficiency

SHG conversion. Photonic crystal defect cavities employ high-Q resonant modes to confine light within ultrasmall volumes, dramatically enhancing light-matter interactions and enabling efficient nonlinear processes in compact structures. Meanwhile, harmonic nanoparticles, benefiting from localized resonance effects, exhibit strong field enhancement and broad spectral range of nonlinear responses, making them highly effective for nanoscale SHG generation. By integrating these advanced out-of-plane architectures, nonlinear photonic devices achieve unprecedented miniaturization while maintaining high performance, paving the way for ultra-compact and functionally versatile nonlinear optical systems.

3. Conclusion and Outlook

In conclusion, we review the advancements in LNOI-based SHG nanostructures. **Figure 12** summarizes the nonlinear conversion efficiencies across the various platforms discussed in this article. The results for in-plane and out-of-plane devices are shown in red and blue colors, report of efficiency values for in-plane LNOI devices in 2015, the technology has undergone a transition from etched waveguides to poled waveguides, further evolving into microcavities. The efficiency has improved from between 10^{-8} and 10^{-4} to above 10^{-1} . Although out-of-plane device was introduced as early as 2013, their efficiency remained lower than that of comparable III-V semiconductor devices (indicated by the dashed line region), due to limitations in the fabrication process until 2022. Today, the efficiency of out-of-plane devices has increased from between 10^{-8} and 10^{-6} to over 10^{-3} . Additionally, in-plane devices predominantly occupy the right side of the figure, exhibiting orders of magnitude higher normalized efficiency compared to the out-of-plane devices, which are mainly clustered on the left. Data points in the upper right corner reflect both higher absolute and normalized efficiency, demonstrating the strong potential of in-plane devices for frequency conversion of low-power lasers.

It is important to note that while in-plane LNOI SHG devices can achieve efficiencies exceeding 50%, their high Q-factors (on the order of 10^5 to 10^6) or narrow PM spectral windows make them highly selective to the incident pump wavelengths. This

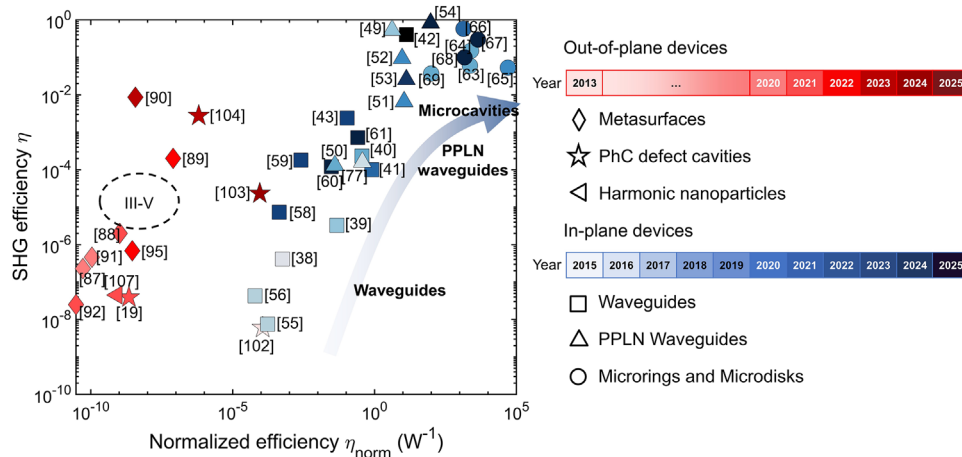


Figure 12. Absolute efficiency (η) and normalized efficiency (η_{norm}) for different LNOI devices.

restricts coupling only to ultra-narrow laser spectral width, proportional to $1/Q$. Furthermore, the high dispersion associated with these high- Q resonances or waveguide modes makes them less suitable for ultrashort pulsed lasers, unless specifically designed for such applications. As a result, most reported in-plane LNOI SHG devices have been demonstrated using monochromatic continuous-wave laser pumps for narrowband SHG. Additionally, the thermo-optic effect of LN allows for thermal tuning of the SHG wavelengths.^[39,43] Beyond narrowband applications, broadband SHG has also been achieved through sophisticated dispersion engineering and mode hybridization in PPLN waveguides, significantly enhancing SHG bandwidth and enabling efficient SHG under femtosecond pulse excitation.^[110,111] In contrast, out-of-plane devices, with Q -factors on the order of 10^2 to 10^3 . Their relatively lower Q -factors, compared to in-plane devices, offer greater flexibility for applications requiring a broader spectral bandwidth, making them better suited for coupling with ultrashort pulsed lasers, including femtosecond pulses. While it is expected that increasing the Q -factor in out-of-plane devices can enhance field localization and nonlinear interactions, it also leads to narrower resonances, which in turn reduces the spectral bandwidth available for SHG.

While the in-plane LN platform has already demonstrated significant success in achieving high SHG efficiency, several challenges remain. Current optical losses during coupling of laser onto optical chip are still a major limitation, and various methods have been proposed to improve coupling efficiency.^[112,113] Moreover, the relatively large footprint of in-plane devices, such as waveguides typically extending millimeters in length and microrings or microdisks with diameters of hundreds of micrometers, presents a requirement for further optimization to reduce the device size. Furthermore, the use of continuous-wave lasers raises concerns about thermal stability, necessitating robust thermal management strategies to ensure consistent and reliable performance.

For out-of-plane devices, the most pressing challenge is to improve their SHG efficiency to be comparable to the in-plane devices. The SHG efficiency can be expressed as^[18,83]

$$\eta \propto Q_r^2 Q_{\text{nr}} \quad (6)$$

However, most current research on metasurfaces has primarily focused on singly resonant regimes at the FF, without fully exploring the potential benefits of SH resonances.^[114,115] For substantial performance improvements, future metasurface designs should aim for dual resonance at both FF and SH frequencies and efficient mode matching between the two. While this has been explored in other materials and systems,^[116–118] it has been scarcely investigated on the LN platform.^[104,119] Additionally, the integration of high- Q resonances with pulsed light presents distinct challenges, particularly in ensuring spectral overlap between the resonance and the pump light.^[120–122] In the perturbative regime, SHG efficiency was reported to reach its maximum when the pulse duration τ and resonance width γ satisfy the condition $\gamma\tau = 5.6$.^[121] Regarding to the total Q -factor of resonance, it comprises both the radiative (Q_r) and non-radiative (Q_{nr}) components, as expressed by $1/Q = 1/Q_r + 1/Q_{\text{nr}}$.^[123] Radiative losses are associated with inherent mode leakage, while non-radiative losses stem from factors such as optical absorption, surface roughness, structural disorder, et al.^[123] Therefore, improving the Q -factor primarily involves a combination of strategies: optimizing the resonance modes to improve Q_r and reducing fabrication-induced imperfections to increase Q_{nr} . For example, by carefully designing the structural asymmetry parameters, a high Q -factor with low radiative loss can be achieved in quasi-BICs.^[124] Additionally, vertical confinement can be enhanced by placing a DBR below the structure, effectively suppressing downward radiation leakage and further improving mode confinement.^[125] To reduce non-radiative losses, using a freestanding membrane can mitigate optical losses introduced by ion contamination during the FIB milling.^[89] In addition, subsequent wet treatments in HF or KOH solutions have proven effective in further improving the Q -factor.^[6,126–128] Notably, while a high Q -factor is often desirable, it is not the sole determining factor for efficient SHG. Optimal nonlinear efficiency is reported to be achieved under the critical coupling condition, where Q_r equals Q_{nr} .^[123] Without these considerations, incident energy cannot be efficiently coupled into the resonance mode, severely limiting performance. Finally, in addition to the single-layer metasurfaces, stacked metasurfaces can be created by stacking multiple layers of nanostructured metasurfaces. They are achieved either by

sequential fabrication, where nonlinear metasurfaces and spacer layers are alternately produced,^[129,130] or by nanoprobe-assisted stacking, where pre-fabricated nonlinear metasurfaces are precisely assembled.^[14] The interaction between the layers allows for more complex behaviors than single-layer metasurfaces. Each layer can have distinct properties, such as different optical responses or phase shifts, which can be tailored independently. Such nonlinear metasurfaces offer novel advantages in flexible dispersion design over conventional phase-matching techniques that rely on intrinsic material properties.^[129] This approach leverages both resonant localized field enhancement and phase engineering simultaneously. Specifically, controlling the geometry and spacing between layers, one can design phase-matched devices with arbitrary transverse phase profiles, enabling the fabrication of advanced nonlinear devices such as metalenses and holographic systems. With these ongoing developments and optimizations, we anticipate that the out-of-plane platform will show remarkable advancements in the near future.

While our primary focus has been on SHG, the implications of our findings extend to other nonlinear phenomena, including sum-frequency generation, third-harmonic generation, four-wave mixing, and optical parametric oscillations.^[131–136] The LN nanostructures discussed in this work show great potential as platforms for developing ultra-compact and efficient nonlinear light sources. These advances open up a broad range of applications, from classical photonics to emerging areas in quantum photonics.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

cavity, lithium niobate on insulator, resonance, second harmonic generation

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